



126 Company of Heros



131 The Sketchpad Whiz



148 Oh,

Digital Leisure

Technology Beyond Work

Touched By Tech

We Want You!

Run a screen saver, help humanity!

Samir Makwana

Normally, we use this space to tell you what people or companies have done for society. This time, we're here to tell you about how you can help society, without even lifting a finger!

Too many of us have the habit of leaving our computers unattended and unused for a while—sometimes for a few minutes, and at times, for hours (that's where all the screensavers come in!). And many of us—especially those of us in offices—have the privilege of an always-on Internet connection. Instead of shutting down and then rebooting, how about assigning your computer's idle time to a productive purpose—like aiding a cancer research?

From Years To Months

In scientific research, huge amounts of processing power—measured in gigaflops, teraflops and even petaflops—are required to perform long and complex calculations in the areas of human protein folding, cryptographic ciphers, and such. These computational tasks are typically performed using supercomputers or other highly powerful computer systems.

Some research projects, such as those trying to find cures for diseases such as AIDS and cancer, can take quite a few years to complete, depending upon the computation they require:

they tend not to have a specified time to completion. But a solution has evolved to cut down on that time. That solution is called *Distributed Computing*.

In distributed computing projects, people contribute their idle computer processing cycles using a middleware platform for the projects open for public participation. Middleware refers to

developments: on the one hand, minicomputers, workstations, and then personal computers; and on the other, computer networks, including, eventually, the Internet. The first Internet-based distributed computing project was started by the DEC System Research Center at Palo Alto, California. The project sent tasks to volunteers via e-mail, who would run these programs in their computers' idle time, then send the results back to DEC—and then get a new task. The DEC System Research Center was taken over by Compaq in 1998 and was merged with the HP SRC Classic lab.

The Distributed.net project, started in 1997, actually made use of the Internet to distribute data for calculation and collect the results. They used independently-owned computers, and allowed the users to download programs that would utilise the idle time of the CPUs. The project was focused on breaking real-world cryptographic ciphers.

[www.distributed.net/projects.php]

The SETI@home (Search for Extra-Terrestrial Intelligence at home) project was initiated in May 1999, hosted by the Space Sciences Laboratory at the University of California at Berkeley. This was the project that really popularised the concept of distributed computing, and you've probably heard of it. SETI@home searches for evidence of extra-terrestrial radio signal transmissions and



Photograph Jiten Gandhi

the software that performs the function of agents that use the CPU cycles on the computer processing the data, and sends the results back to the server.

The Paradigm

Distributed computing goes back a long way, actually. It began in 1970 with the emergence of two